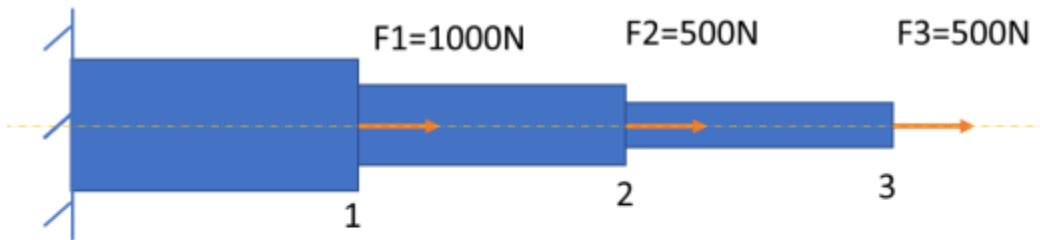


Q1. Figure 1 shows a liner-elastic three-bar system. Please use Direct Stiffness Method to

1) Derive the element stiffness matrix. 20 points

2) Calculate the displacements at the end of each bar and fill the values into the table: 30 points

Node	1	2	3
Displacement (U_mag). m	$1 \cdot 10^{-8}$	$3 \cdot 10^{-8}$	$4.333 \cdot 10^{-8}$



$E1=200\text{GPa}$

Poisson ratio = 0.3

$A1=1\text{ m}^2$

$L1 = 1\text{m}$

$E2=100\text{GPa}$

Poisson ratio = 0.3

$A2=0.5\text{m}^2$

$L2 = 1\text{m}$

$E3=150\text{GPa}$

Poisson ratio = 0.3

$A3=0.25\text{ m}^2$

$L3 = 1\text{m}$

$$\frac{A_1 E_1}{L_1} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{Bmatrix} F_1 \\ F_2 \end{Bmatrix} \rightarrow \frac{A_1 E_1}{L_1} = \frac{200 \cdot 10^9 \text{ Pa} \cdot 1 \text{ m}^2}{1 \text{ m}} = 200 \cdot 10^9$$

$$\frac{A_2 E_2}{L_2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} u_2 \\ u_3 \end{Bmatrix} = \begin{Bmatrix} F_2 \\ F_3 \end{Bmatrix} \rightarrow \frac{A_2 E_2}{L_2} = \frac{0.5 \cdot 100 \cdot 10^9}{1} = 50 \cdot 10^9$$

$$\frac{A_3 E_3}{L_3} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} u_3 \\ u_4 \end{Bmatrix} = \begin{Bmatrix} F_3 \\ F_4 \end{Bmatrix} \rightarrow \frac{A_3 E_3}{L_3} = \frac{0.25 \cdot 150 \cdot 10^9}{1} = 37.5 \cdot 10^9$$

$$10^9 * \begin{pmatrix} 200 & -200 & 0 & 0 \\ -200 & 200 + 50 & -50 & 0 \\ 0 & -50 & 50 + 37.5 & -37.5 \\ 0 & 0 & -37.5 & 37.5 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 1000 \\ 500 \\ 500 \end{pmatrix}$$

Cross these out because Node 1 will not move because of the Boundary condition keeping it in the same location.

$$10^9 \begin{pmatrix} 200 & -200 & 0 & 0 \\ -200 & 250 & -50 & 0 \\ 0 & -50 & 87.5 & -37.5 \\ 0 & 0 & -37.5 & 37.5 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 1000 \\ 500 \\ 500 \end{pmatrix}$$

$$10^9 \begin{pmatrix} 250 & -50 & 0 \\ -50 & 87.5 & -37.5 \\ 0 & -37.5 & 37.5 \end{pmatrix} \begin{pmatrix} u_2 \\ u_3 \\ u_4 \end{pmatrix} = \begin{pmatrix} 1000 \\ 500 \\ 500 \end{pmatrix} \text{ Element stiffness matrix}$$

Using trivial MATLAB code to solve this matrix we can find that the results are:

```
A=[250,-50,0;-50,87.5,-37.5;0,-37.5,37.5];
B=[1000;500;500];
x=A\B
```

$$10^9 \begin{bmatrix} 250 & -50 & 0 \\ -50 & 87.5 & -37.5 \\ 0 & -37.5 & 37.5 \end{bmatrix} \begin{Bmatrix} u_2 \\ u_3 \\ u_4 \end{Bmatrix} = \begin{Bmatrix} 1000 \\ 500 \\ 500 \end{Bmatrix} \rightarrow \begin{Bmatrix} u_2 \\ u_3 \\ u_4 \end{Bmatrix} = \begin{bmatrix} \left(\frac{10}{10^9}\right) \\ \left(\frac{30}{10^9}\right) \\ \left(\frac{43.333}{10^9}\right) \end{bmatrix}$$

Verifying our results with ABAQUS, we get:

Node Label	U.Magnitude
@Loc 1	

1	0.
2	10.000E-9
3	30.000E-9
4	43.333E-9

Q2. Please build a FE model to analyze the stress distribution and displacement in the following truss setup ($E=2.0E+11\text{Pa}$, $A=1\text{m}^2$, Poisson ratio=0.3):

1) Create a short report in PDF or Word format to display the Von Mises stress and the magnitude of displacements 20 points

2) Extract the displacement values for N1, N2 and N3 and fill in the following table: 20 points

Node	U_mag, m	U_1, m	U_2, m
N1	2.77565E-09	915.063E-12	2.62047E-09
N2	1.83013E-09	1.83013E-09	-183.012E-36
N3	7.51075E-09	7.03878E-09	2.62047E-09

3) Save and upload the FE model

10 points

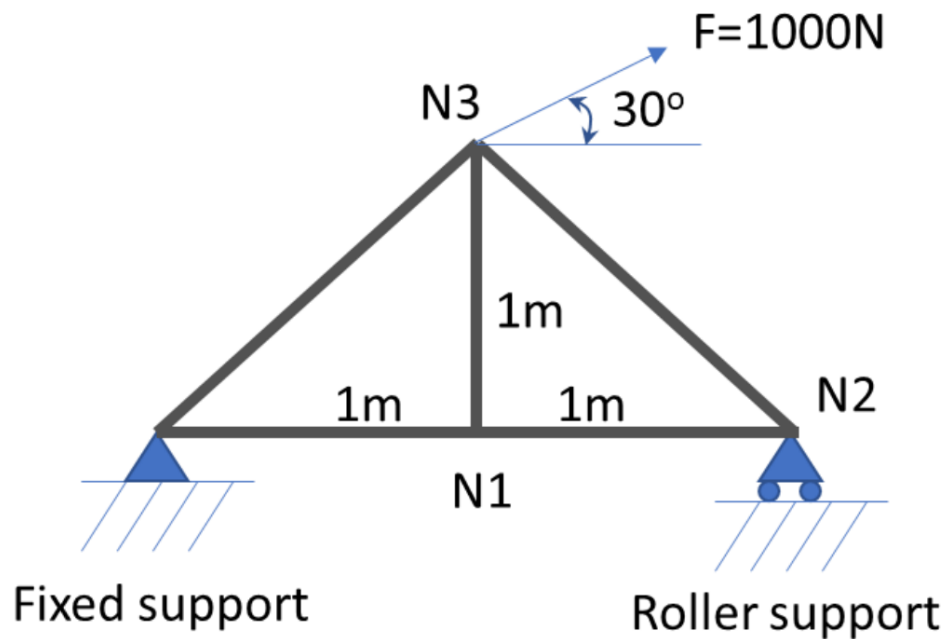


Figure 2.

Mises and Magnitude:

Field Output Report, written Sat Feb 5 13:41:35 2022

Source 1

ODB: C:/Users/User/Desktop/HW1P2/HW1P2Job2.odb

Step: ApplyLoadAndBC

Frame: Increment 1: Step Time = 1.000

Node Label	U.Magnitude	S.Mises
@Loc 1	@Loc 1	

1	2.77565E-09	122.008
2	1.83013E-09	220.916
3	7.51075E-09	408.248